

LEAP 2025 Mathematics

2018 Practice Test

Grade 4

Session 1

Directions:

Today, you will take Session 1 of the Grade 4 Mathematics Test. You will not be able to use a calculator in this session.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test booklet. Do not make any stray pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

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GO ON ►

1. Which expression is equivalent to $6 \times \frac{2}{3}$?

(A) $3 \times \frac{1}{3}$

(B) $4 \times \frac{1}{3}$

(C) $8 \times \frac{1}{3}$

(D) $12 \times \frac{1}{3}$

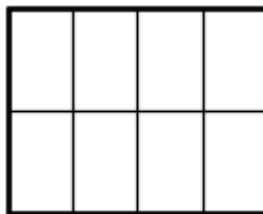
2. Casey spent 18 minutes coloring. She spent 6 times as many minutes reading.

How much time, in minutes, did Casey spend reading?

Enter your answer in the box.

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

3. The rectangle is divided into eight equal sections.



Jodi colors 4 sections. Then she colors 3 more sections.

Which **two** of these represent the fraction of the rectangle that Jodi colors in all?

Select the **two** correct answers.

(A) $\frac{4}{8} + \frac{3}{8}$

(B) $4 + 3$

(C) $\frac{8}{4} + \frac{8}{3}$

(D) $\frac{1}{8} + 3$

(E) $\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$

4. Enter your answer in the box.

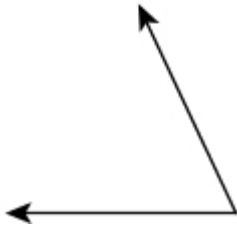
$5,314 - 4,983 =$

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

5. Which angle has a measure of 65° ?

You can use a protractor to help you find the answer.

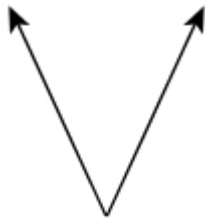
(A)



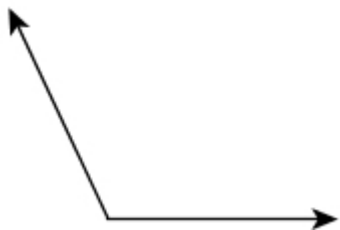
(B)



(C)



(D)

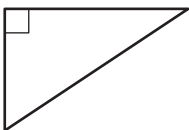
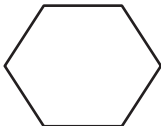


6. What is the expanded form of 50,210?

- (A) $5,000 + 20 + 1$
- (B) $5,000 + 200 + 10$
- (C) $50,000 + 20 + 1$
- (D) $50,000 + 200 + 10$

GO ON ►

7. Which **three** shapes appear to have at least two parallel sides?

- (A) 
- (B) 
- (C) 
- (D) 
- (E) 

8. Ryan makes 6 backpacks. He uses $\frac{3}{4}$ yard of cloth to make each backpack. What is the total amount of cloth, in yards, Ryan uses to make all 6 backpacks?

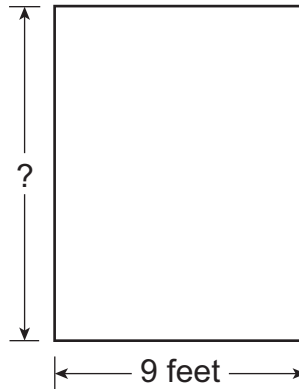
Ⓐ $1\frac{1}{2}$

Ⓑ $2\frac{1}{4}$

Ⓒ $4\frac{1}{2}$

Ⓓ $6\frac{3}{4}$

9. The area of the rectangular sandbox at Dave's school is 108 square feet.
The sandbox has a width of 9 feet, as shown in the diagram.



What is the length, in feet, of the sandbox?

Enter your answer in the box.

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

10. Which **two** equations represent the statement “48 is 6 times as many as 8”?

Select the **two** correct answers.

☐ Ⓐ $48 = 6 + 8$

☐ Ⓑ $48 = 6 \times 8$

☐ Ⓒ $48 = 6 \times 6$

☐ Ⓓ $48 = 8 + 6$

☐ Ⓔ $48 = 8 \times 6$

11. Use the information provided to answer Part A and Part B for question 11.

Jordan places two boards end to end to make one shelf. The first board is $\frac{47}{100}$ meter long. The second board is $\frac{5}{10}$ meter long.

Part A

What fraction is equivalent to $\frac{5}{10}$ and has a denominator of 100?

- (A) $\frac{5}{100}$
- (B) $\frac{50}{100}$
- (C) $\frac{105}{100}$
- (D) $\frac{150}{100}$

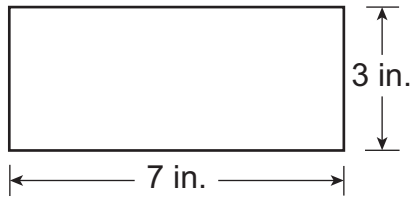
Part B

What is the total length, in meters, of the two boards?

- (A) $9\frac{7}{10}$
- (B) $5\frac{2}{10}$
- (C) $\frac{97}{100}$
- (D) $\frac{52}{100}$

Use the information provided to answer Part A and Part B for question 12.

12. A rectangle is shown.



GO ON ►

Part A

A student uses square tiles measuring 1 inch on each side to find the area of the rectangle. Her reasoning is shown.

I covered the top and bottom edges of the rectangle with 7 tiles each.



I then covered the left and right edges with 3 tiles each. I added up all the tiles I used to get a total area of 20 square inches. $7 + 7 + 3 + 3 = 20$.



Identify the two errors in the student's reasoning and describe how to correctly use square tiles to find the area of the rectangle. Give the correct area of the rectangle.

Enter your answers and your description in the box provided.

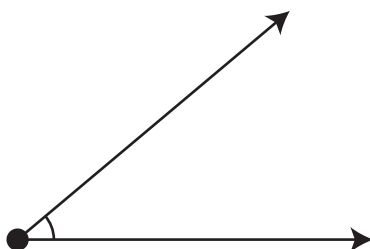
GO ON ►

Part B

Write a multiplication sentence that models how to find the area of the rectangle shown.

Enter your multiplication sentence in the box provided.

13. Look at the angle shown.



Which measure is closest to the measure of the angle?

- (A) 140°
- (B) 90°
- (C) 40°
- (D) 15°

14. Part A

A school's art teacher needs 200 sticks of clay. An art shop donates 9 small boxes of clay and 6 large boxes of clay.

Box Size	Number of Sticks of Clay in Each Box
small	7
large	10

How many more sticks of clay will the art teacher need?

Enter your answer in the box.

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

GO ON ►

Part B

The art teacher buys the rest of the clay he needs in large boxes. The cost of 1 large box of clay is \$14. What is the total cost for these boxes of clay? Show or explain your work.

Enter your answer and your work or explanation in the box provided.

15. Enter your answer in the box to make the number sentence true.

$$5,039 \times 8 = \boxed{}$$

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9



Session 2

Directions:

Today, you will take Session 2 of the Grade 4 Mathematics Test. You will not be able to use a calculator in this session.

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GO ON ►

16. An animal weighs 4 pounds. A bald eagle weighs 3 times as much as this animal. How many pounds does the bald eagle weigh?

Enter your answer in the box.

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

17. Which **three** comparisons are correct?

- (A) 0.4 meter > 0.04 meter
- (B) 0.04 meter > 0.3 meter
- (C) 0.3 meter < 0.5 meter
- (D) 0.5 meter > 0.65 meter
- (E) 0.65 meter > 0.61 meter
- (F) 0.65 meter < 0.04 meter

18. Divide 738 by 6.

Enter your answer in the box.

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

GO ON ►

Use the information provided to answer Part A and Part B for question 19.

19. Each student in a class chose one sport to play. The table shows the fractions of all students who chose each sport.

Sport	Fraction of All Students
soccer	$\frac{3}{10}$
football	$\frac{2}{10}$
hockey	$\frac{1}{10}$
basketball	$\frac{4}{10}$

Part A

Which equation can be used to find s , the fraction of all students that chose to play either soccer or basketball?

- (A) $\frac{3}{10} + \frac{4}{10} = s$
- (B) $\frac{2}{10} - \frac{1}{10} = s$
- (C) $\frac{4}{10} + \frac{2}{10} = s$
- (D) $\frac{4}{10} - \frac{3}{10} = s$

Part B

What fraction of all the students chose to play either soccer or basketball?

(A) $\frac{1}{10}$

(B) $\frac{3}{10}$

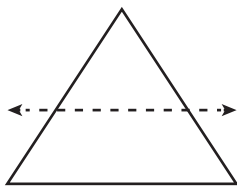
(C) $\frac{6}{10}$

(D) $\frac{7}{10}$

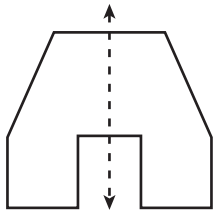
20. Which of these show lines of symmetry?

Select the **three** correct answers.

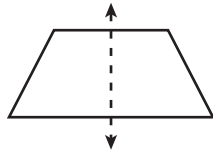
(A)



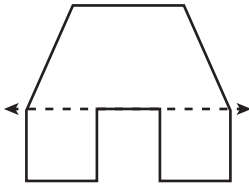
(B)



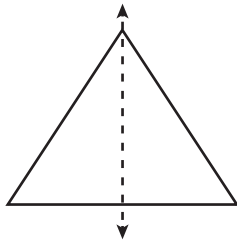
(C)



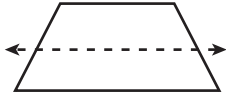
(D)



(E)



(F)



GO ON ►

21. What is the value of $9,348 + 2,237$?

Enter your answer in the box.

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

GO ON ►

Use the information provided to answer Part A and Part B for question 22.

22. Of the students in one school, $\frac{1}{12}$ play soccer, $\frac{3}{8}$ play basketball, $\frac{2}{5}$ take music lessons, and $\frac{2}{6}$ take dance lessons.

Part A

Which fraction is equivalent to the fraction of students who take music lessons at the school?

- (A) $\frac{3}{6}$
- (B) $\frac{5}{8}$
- (C) $\frac{4}{10}$
- (D) $\frac{4}{12}$

Part B

Which list orders the fractions from least to greatest?

- (A) $\frac{1}{12}, \frac{2}{5}, \frac{2}{6}, \frac{3}{8}$
- (B) $\frac{2}{5}, \frac{3}{8}, \frac{2}{6}, \frac{1}{12}$
- (C) $\frac{2}{5}, \frac{2}{6}, \frac{3}{8}, \frac{1}{12}$
- (D) $\frac{1}{12}, \frac{2}{6}, \frac{3}{8}, \frac{2}{5}$

23. Which of these numbers are prime numbers?

Select the **three** numbers that are prime.

- (A) 15
- (B) 19
- (C) 27
- (D) 37
- (E) 43
- (F) 51

24. Mr. Kowolski ordered 35 boxes of granola bars. Each box contained 24 granola bars.

What is the total number of granola bars Mr. Kowolski ordered?

Enter your answer in the box.

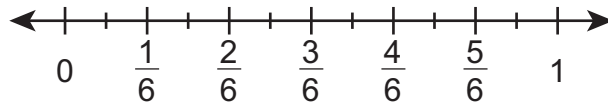
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

25. The value of the digit 4 in the number 42,780 is 10 times the value of the digit 4 in which number?

- (A) 34,651
- (B) 146,703
- (C) 426,135
- (D) 510,400

26. Explain how to find $2 \times \frac{5}{12}$ using the number line.

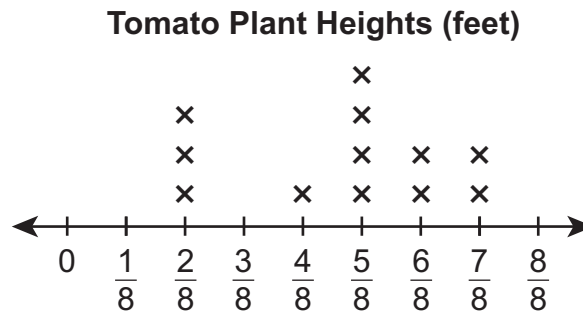
Find the product.



Enter your answer and your explanation in the box provided.

GO ON ►

27. The line plot represents the heights, in feet, of tomato plants in a garden.



What is the difference, in feet, between the tallest and shortest plant heights?

- (A) $\frac{1}{8}$
 - (B) $\frac{3}{8}$
 - (C) $\frac{5}{8}$
 - (D) $\frac{7}{8}$
28. Which statement about angles is true?
- (A) An angle is formed by two rays that do not have the same endpoint.
 - (B) An angle that turns through $\frac{1}{360}$ of a circle has a measure of 360 degrees.
 - (C) An angle that turns through five 1-degree angles has a measure of 5 degrees.
 - (D) An angle measure is equal to the total length of the two rays that form the angle.

29. A student uses tubes of paint to draw on 1 poster and 2 shirts.

- The student uses 6 tubes of paint to draw on the poster.
- The number of tubes used for the poster is 3 times the number of tubes used for each shirt.
- Each tube contains $\frac{1}{3}$ ounce of paint.

How many ounces of paint does the student use for 1 shirt? How many ounces of paint does the student use to make 1 poster and 2 shirts? Show your work or explain your answers.

Enter your answers and your work or explanation in the box provided.



Session 3

Directions:

Today, you will take Session 3 of the Grade 4 Mathematics Test. You will not be able to use a calculator in this session.

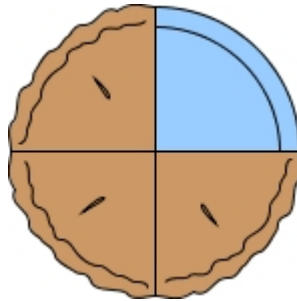
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GO ON ►

30. Jasmine ate $\frac{1}{4}$ of a pie. She drew a model to represent the fraction of the pie that she ate.



Which fraction is equivalent to the fraction of the pie that Jasmine ate?

- (A) $\frac{2}{5}$
- (B) $\frac{3}{6}$
- (C) $\frac{2}{8}$
- (D) $\frac{1}{12}$

31. The Amazon River is about 6,516 kilometers long.

The Mississippi River is about 3,775 kilometers long.

What is the difference, in kilometers, between these two lengths?

Enter your answer in the box.

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

32. Which expressions have a value of $\frac{8}{12}$?

Select the **three** correct answers.

- (A) $\frac{2}{12} + \frac{4}{12}$
- (B) $\frac{6}{12} + \frac{2}{12}$
- (C) $\frac{1}{12} + \frac{2}{12} + \frac{4}{12}$
- (D) $\frac{3}{12} + \frac{2}{12} + \frac{2}{12} + \frac{1}{12}$
- (E) $\frac{1}{12} + \frac{2}{12} + \frac{1}{12} + \frac{2}{12} + \frac{1}{12} + \frac{2}{12}$
- (F) $\frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12}$

33. Which numbers make the comparison true?

$$27,768 < \square$$

Select the **two** correct answers.

- (A) 27,759
- (B) 28,744
- (C) 26,773
- (D) 27,568
- (E) 27,836

34. The length of a desktop is 4 feet. How many inches is the length of the desktop?

Enter your answer in the box.

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

35. During a class trip to an apple farm, a group of students picked 2,436 apples. They packed them into 6 boxes to take to the local food bank. If each box held the same number of apples, how many apples were in each box?

- (A) 46 apples
- (B) 406 apples
- (C) 460 apples
- (D) 4,060 apples

36. What is the decimal form of the fraction $\frac{9}{100}$?

- (A) 0.9
- (B) 0.09
- (C) 0.009
- (D) 0.0009

37. Enter your answer in the box.

$$3,950 + 405 =$$

•	•	•	•	•	•
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

38. Isabel used $\frac{2}{3}$ cup of strawberries in a fruit salad. She used less than $\frac{2}{3}$ cup of blueberries in the same salad. Which of the following could be the fraction of a cup of blueberries that Isabel used?

Select the **three** fractions that could represent the fraction of a cup of blueberries.

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{4}{5}$

(D) $\frac{5}{6}$

(E) $\frac{3}{8}$

GO ON ►

39. Select the **three** choices that are factor pairs for the number 28.

- (A) 1 and 28
- (B) 2 and 14
- (C) 3 and 9
- (D) 4 and 7
- (E) 6 and 5
- (F) 8 and 3

40. In one year, Janie sent 4,368 text messages. Tanner sent 4 times as many text messages as Janie. How many more text messages did Tanner send than Janie?

Enter your answer in the box.

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

41. A student's work to add the mixed numbers $1\frac{3}{4}$ and $2\frac{3}{4}$ is shown.

$$\begin{aligned}1\frac{3}{4} + 2\frac{3}{4} &= \frac{4}{4} + \frac{3}{4} + \frac{8}{4} + \frac{3}{4} \\&= \frac{4 + 3 + 8 + 3}{4 + 4 + 4 + 4} \\&= \frac{18}{16}\end{aligned}$$

Explain any errors you see in the work. Find the correct solution. Show your work or explain your answer.

Enter your explanation, your solution, and your work or explanation in the box provided.

42. A photographer has 591 photos of animals and 234 photos of plants. He wants to put all of the photos into photo books. Each page of the photo books holds 8 photos. What is the fewest number of pages he could use in the photo books?
- Ⓐ 73
 - Ⓑ 74
 - Ⓒ 103
 - Ⓓ 104

Use the information provided to answer Part A and Part B for question 43.

43. Carl is training for a bike race.

- On Thursday, he rides his bike 7 miles.
- On Friday, he rides his bike 2 times the number of miles he rides on Thursday.
- On Saturday, he rides his bike 9 miles.

Part A

Explain how to find the total number of miles Carl rides his bike on Thursday, Friday, and Saturday. Include the total number of miles he rides in your explanation.

Enter your explanation in the box provided.

GO ON ►

Part B

Carl wants to ride his bike a total of 36 miles over the next three days. He will add the same number of miles to each distance from Part A.

Show or explain how to find the number of miles Carl should add to his distance each day.

- Include the number of miles added to his distance each day in your work or explanation.
- Include the new distance for each of the three days in your work or explanation.

Enter your explanation or your work in the box provided.



This document contains the answer keys and rubrics for the LEAP 2025 Grade 4 Mathematics Paper-Based Practice Test.

Session 1				
Task #	Task Type	Value (points)	Key	Alignment
1	I	1	D	4.NF.B.4b
2	I	1	108	4.OA.A.2
3	I	1	A, E	4.NF.B.3a
4	I	1	331	4.NBT.B.4
5	I	1	A	4.MD.C.6
6	I	1	D	4.NBT.A.2
7	I	1	B, D, E	4.G.A.2
8	I	1	C	4.NF.B.4c
9	I	1	12	4.MD.A.3
10	I	1	B, E	4.OA.A.1
11	I	2	Part A: B Part B: C	4.NF.C.5
12	II	4	Part A: rubric Part B: rubric	LEAP.II.4.5 (3.MD.C)
13	I	1	C	4.MD.C.6
14	III	3	Part A: 77 Part B: rubric	LEAP.III.4.1 (4.OA.A.3, 4.NBT.B.4)
15	I	1	40,312	4.NBT.B.5

Session 2				
Task #	Task Type	Value (points)	Key	Alignment
16	I	1	12	4.OA.A.2
17	I	1	A, C, E	4.NF.C.7
18	I	1	123	4.NBT.B.6
19	I	2	Part A: A Part B: D	4.NF.B.3d
20	I	1	BCE	4.G.A.3
21	I	1	11,585	4.NBT.B.4
22	I	2	Part A: C Part B: D	LEAP.I.4.1 (4.NF.A)
23	I	1	B, D, E	4.OA.B.4d
24	I	1	840	LEAP.I.4.2 (4.NBT.B.5, 4.OA.A)

Session 2				
Task #	Task Type	Value (points)	Key	Alignment
25	I	1	A	4.NBT.A.1
26	II	3	rubric	LEAP.II.4.7 (4.NF.B.4a, 4.NF.B.4b)
27	I	1	C	4.MD.B.4
28	I	1	C	4.MD.C.5c
29	III	3	rubric	LEAP.III.4.1 (4.OA.A.2, 4.NF.B.4c)

Session 3				
Task #	Task Type	Value (points)	Key	Alignment
30	I	1	C	4.NF.A.1
31	I	1	2741	LEAP.I.4.7 (4.NBT.B.4, 4.OA.A)
32	I	1	B, D, F	4.NF.B.3b
33	I	1	B, E	4.NBT.A.2
34	I	1	48	4.MD.A.1
35	I	1	B	LEAP.I.4.4 (4.NBT.B.6, 4.OA.A)
36	I	1	B	4.NF.C.6
37	I	1	4,355	4.NBT.B.4
38	I	1	A, B, E	4.NF.A.2
39	I	1	A, B, D	4.OA.B.4a
40	I	1	13,104	4.OA.A.3
41	II	3	rubric	LEAP.II.4.6 (4.NF.B.3c)
42	I	1	D	4.OA.A.3
43	III	6	Part A: rubric Part B: rubric	LEAP.III.4.2 (3.OA.A, 3.OA.D.8)

RUBRICS

Task # 12	
Part A	
Score	Description
3	Student response includes the following 3 elements: • Reasoning component: 3 points ○ Identifies error of overlapping tiles on corners ○ Identifies error of not completely covering the rectangle with tiles ○ Explains a way to correctly cover the rectangle and determines the area of the rectangle is 21 square inches Sample Student Response: One error the student made was she covered each corner of the rectangle twice. Another error she made was she didn't completely cover the entire rectangle. To correctly determine the area, you should cover the entire rectangle with squares without overlapping. If I do this, I would cover the top and bottom edges with 7 tiles each, then I could add another 7 tiles to cover the middle section of the rectangle. In all, I used 21 tiles to cover the entire rectangle with no overlaps. This means that the area of the rectangle is 21 square inches. $7 + 7 + 7 = 21$ Or other valid explanation. Notes: • If the error made is shown or stated as the perimeter is being found, not area, a point can be given for either the first or second element, but not a point for each. • Work that correctly shows the area of the rectangle addresses the requirement to explain the way to cover the rectangle.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.
Part B	
Score	Description
1	Student response includes the following element: • Computation component: 1 point ○ Valid multiplication sentence that shows how to find the area of the rectangle shown Sample Student Response: $7 \times 3 = 21$
0	Student response is incorrect or irrelevant.

Task #14	
Part A	
Score	Description
1	Student response includes the following element: • Computation component: 1 point ○ Correct number of sticks of clay, 77
0	Student response is incorrect or irrelevant.
Part B	
Score	Description
2	Student response includes the following 2 elements: • Modeling component: 1 point ○ Correct work or explanation for how to find the total cost of the remaining boxes of clay • Computation component: 1 point ○ Correct cost, \$112 Sample Student Response: $77 \div 10 = 7, \text{ with a remainder of } 7 \text{ so the teacher needs } 8 \text{ boxes.}$ $8 \times 14 = 112$ \$112 Or other valid explanation.
1	Student response contains 1 of the 2 elements.
0	Student response is incorrect or irrelevant.

Task #26

Score	Description
3	Student response includes the following 3 elements: • Reasoning component: 2 points ○ Correct explanation of how to find $\frac{5}{12}$ using the number line ○ Correct explanation of how to find $2 \times \frac{5}{12}$ using the number line • Computation component: 1 point ○ Correct product, $\frac{10}{12}$ or equivalent Sample Student Response: I know that each tick mark on the number line is equivalent to $\frac{1}{12}$, so to find $\frac{5}{12}$, I would count 5 of the tick marks. Then to find $2 \times \frac{5}{12}$, I would count $\frac{5}{12}$ two times starting at zero on the number line. I would land on $\frac{5}{6}$, which is the same as $\frac{10}{12}$. The product is $\frac{10}{12}$. Or other valid explanation. Note: Student responses must provide explanations to receive the reasoning component points. Simply identifying the locations of $\frac{5}{12}$ and $\frac{10}{12}$ is not sufficient for reasoning credit.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #29

Score	Description
3	Student response includes the following 3 elements: • Computation component: 2 points ○ Correct amount of paint used for 1 shirt, $\frac{2}{3}$ ounce ○ Correct amount of paint used for the poster and 2 shirts, $\frac{10}{3}$ ounces or equivalent • Modeling component: 1 point ○ Correct work or explanation for both the amount of paint used for 1 shirt and for the amount of paint used for the poster and 2 shirts Sample Student Response: I found the number of tubes used for each shirt by figuring out what number times 3 is equal to 6. Since $3 \times 2 = 6$, I now know that for each shirt, 2 tubes are used. To find the number of ounces in 2 tubes, I solved $2 \times \frac{1}{3} = \frac{2}{3}$. To find the number of ounces used for 1 poster, I solved $6 \times \frac{1}{3} = \frac{6}{3}$. For the total number of ounces used for 1 poster and 2 shirts, I added $\frac{2}{3} + \frac{2}{3} + \frac{6}{3} = \frac{10}{3}$. So the total number of ounces used for the poster and 2 shirts is $\frac{10}{3}$. Notes: • Student may earn the computation point for an incorrect amount of paint used for the poster and 2 shirts if it is based on an incorrect amount of paint used for 1 shirt with the correct work shown. • Student may earn the modeling point even if computation is incorrect as long as he or she then shows valid work or explanation.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #41

Score	Description
3	Student response includes the following 3 elements: • Reasoning component: 2 points ○ Correct explanation of errors in work shown ○ Correct work or explanation given for finding correct solution • Computation component: 1 point ○ Correct solution, $\frac{18}{4}$ or equivalent Sample Student Response: The fractions are incorrectly added. The student should not add together the values in the denominator. The correct way to do the problem is: $1\frac{3}{4} + 2\frac{3}{4} = \frac{4}{4} + \frac{3}{4} + \frac{8}{4} + \frac{3}{4}$ $= \frac{4+3+8+3}{4}$ $= \frac{18}{4}$ Or other valid explanation or work.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #43	
Part A	
Score	Description
3	Student response includes the following 3 elements: • Modeling component: 2 points ○ Correct work or explanation to find the total number of miles Carl rides on Friday ○ Correct work or explanation to find the total number of miles Carl rides in three days • Computation component: 1 point ○ Correct answer, 30 Sample Student Response: Carl rides twice as far Friday as he does on Thursday. $7 \times 2 = 14$. Carl rides $7 + 14 + 9 = 30$ miles in all. Notes: • A variety of explanations are possible. As long as the student explains how to find the total number of miles Carl rides on Friday and in the three days, credit should be given. • If a computation mistake is made, credit cannot be given for computation but can be given for a valid explanation.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #43**Part B**

Score	Description
3	Student response includes the following 3 elements: • Modeling component: 2 points ○ Correct work or explanation to find the number of total additional miles Carl must ride ○ Correct work or explanation to find the number of additional miles Carl should ride each day • Computation component: 1 point ○ Correct answer, 2 Sample Student Response: For Carl to ride 36 miles in the next 3 days, he needs to ride $36 - 30 = 6$ more miles. For Carl to ride 6 more miles in 3 days, he must ride $6 \div 3 = 2$ miles farther each day. Notes: • A variety of explanations are valid. As long as the student uses a valid method to solve the problem, credit should be given. • If a computation mistake is made, credit cannot be given for computation but should be given for a valid explanation of how to solve the problem. • If a mistake was made in Part A but carried through Part B correctly, credit can be given for Part B.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.